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Mesh-Controlled SWARM ROBOTS For Thermal Surveys And Analysis

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Mesh swarm robots offer a practical approach to collaborative robotics, with multiple autonomous units communicating, sharing information, and coordinating their actions to perform complex tasks. These systems are particularly suited to applications that require distributed sensing and decision-making, including environmental monitoring, area surveys, gas-leak detection, and search-and-rescue operations.

In this system, each robot in the mesh swarm is equipped with an AMG8833 thermal sensor, N20 motors, motor drivers, and an IndusBoard that integrates multiple onboard sensors. Additional end-effectors, such as claws, can also be fitted as required. The robots operate through a mesh network supporting multiple master, receiver, and bridge nodes. A master node receives high-level commands from a Wi-Fi access point or bridge and distributes tasks among the individual robots, enabling coordinated thermal surveys and analysis.

As the robots move, they continuously exchange position data and sensor readings with one another and report the collected information to the master node. The master aggregates and analyses the data, particularly the thermal readings, and generates alerts when a robot detects a thermal anomaly, fire source, or high-temperature signature that may indicate a gas leak. This coordinated mesh architecture helps improve the speed, coverage, and reliability of thermal inspection and emergency-response missions. Fig. 1 shows the author's prototype. The components required to build this system are listed in the Bill of Materials table.

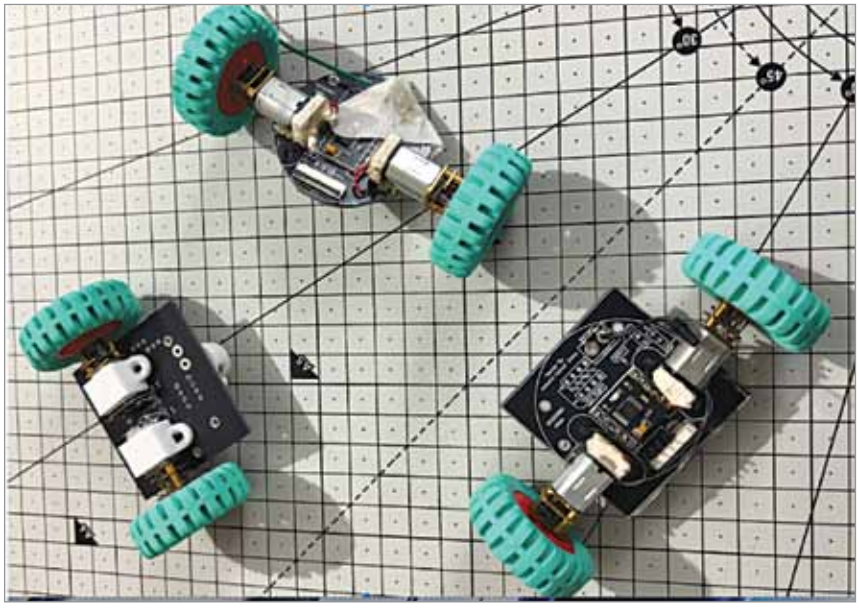


Fig. 1: Author's prototype

Circuit and working

The master and swarm robot nodes use essentially the same circuit, except that the AMG8833 thermal sensor is required only on the swarm robots. The DRV8833 controls the two DC gear motors, while the IndusBoard V2 connects to the main PCB through an FPC connector, with other required sensors and interfaces integrated on the board.

BILL OF MATERIALS

Name	Designator	Description	Qty
Screw terminal	CONN1	2-pin battery connector	4
XW0520012-240R-001	FPC1	FPC SMD 24-pin, 0.50mm pitch connector	4
N20 motor	M1, M2	N20 DC motor, 3V-7V DC	8
AMG8833 module	S1	AMG8833 thermal sensor module	2
IndusBoard V2	IC1	IndusBoard Coin V2	3
DRV8833 module	IC2	DRV8833 dual motor driver module	8
Soldering jumper, 2-pin	—	2-pin soldering jumper	8
Buzzer	PB1	Piezo buzzer	4

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Ready to bring this circuit to life? To unlock the circuit diagram, component list, PCB layout and complete step-by-step build instructions visit: https://efy.co.in/06z1vu